

```
import tensorflow as tf
```

```
from tensorflow.keras import layers, models
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```
import matplotlib.pyplot as plt
```

```
mnist = tf.keras.datasets.mnist
```

```
(x_train, y_train), (x_test, y_test) = mnist.load_data()
```

```
x_train, x_test = x_train / 255.0, x_test / 255.0
```

```
x_train = x_train.reshape(-1, 28, 28, 1)
```

```
x_test = x_test.reshape(-1, 28, 28, 1)
```

```
model = models.Sequential([
```

```
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),
```

```
    layers.MaxPooling2D((2, 2)),
```

```
    layers.Conv2D(64, (3, 3), activation='relu'),
```

```
    layers.MaxPooling2D((2, 2)),
```

```
    layers.Flatten(),
```

```
    layers.Dense(128, activation='relu'),
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```
    layers.Dropout(0.5),
```

```

        layers.Dense(10, activation='softmax') # 10 classes for digits 0-9

    ])

model.compile(optimizer='adam',

              loss='sparse_categorical_crossentropy',

              metrics=['accuracy'])

model.fit(x_train, y_train, epochs=5, validation_data=(x_test, y_test))

test_loss, test_acc = model.evaluate(x_test, y_test)

print(f"Test accuracy: {test_acc:.4f}")

predictions = model.predict(x_test[:5])

for i in range(5):

    plt.imshow(x_test[i].reshape(28, 28), cmap='gray')

    plt.title(f"Predicted: {predictions[i].argmax()} | Actual: {y_test[i]}")

    plt.axis('off')

plt.show()

```